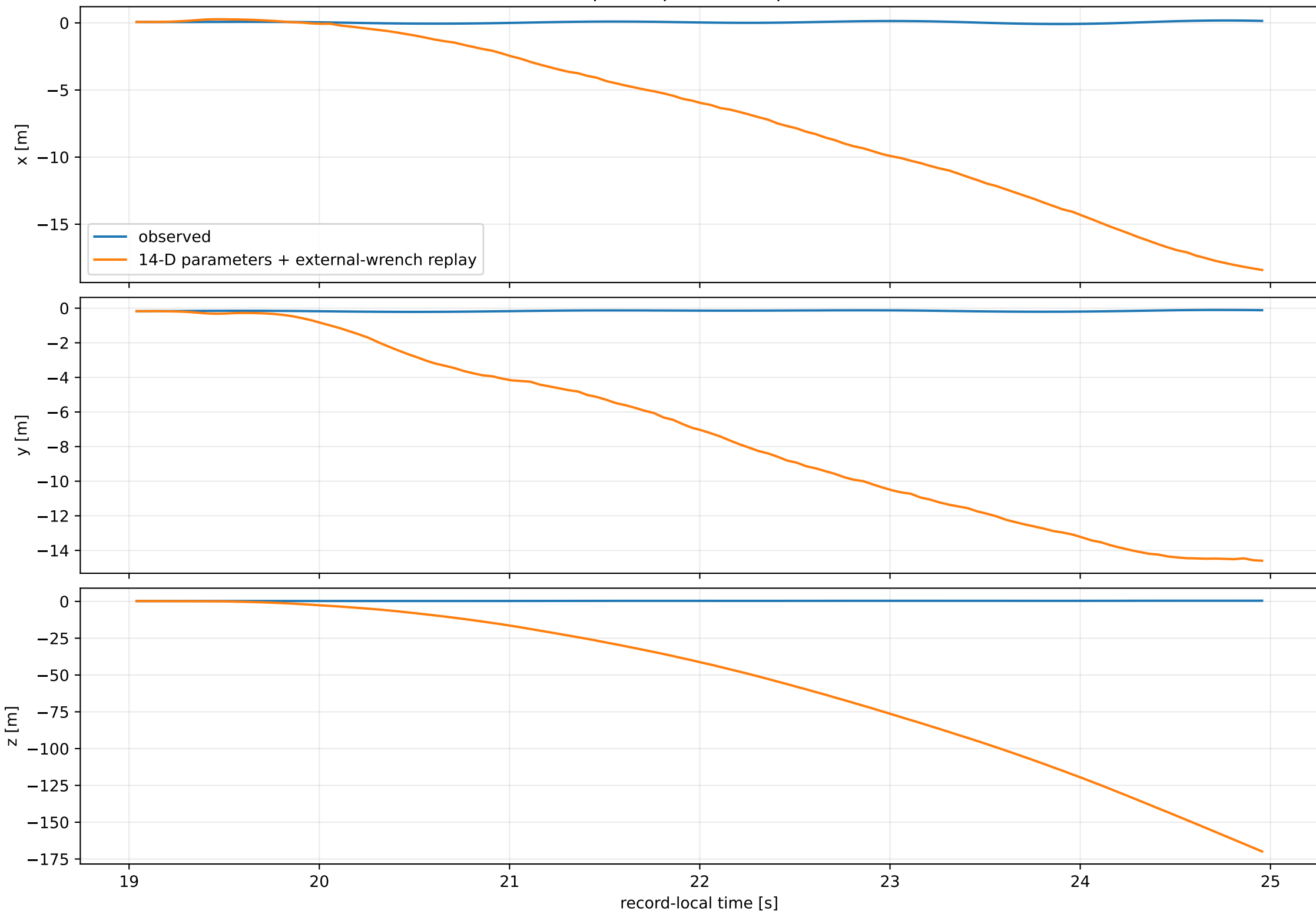
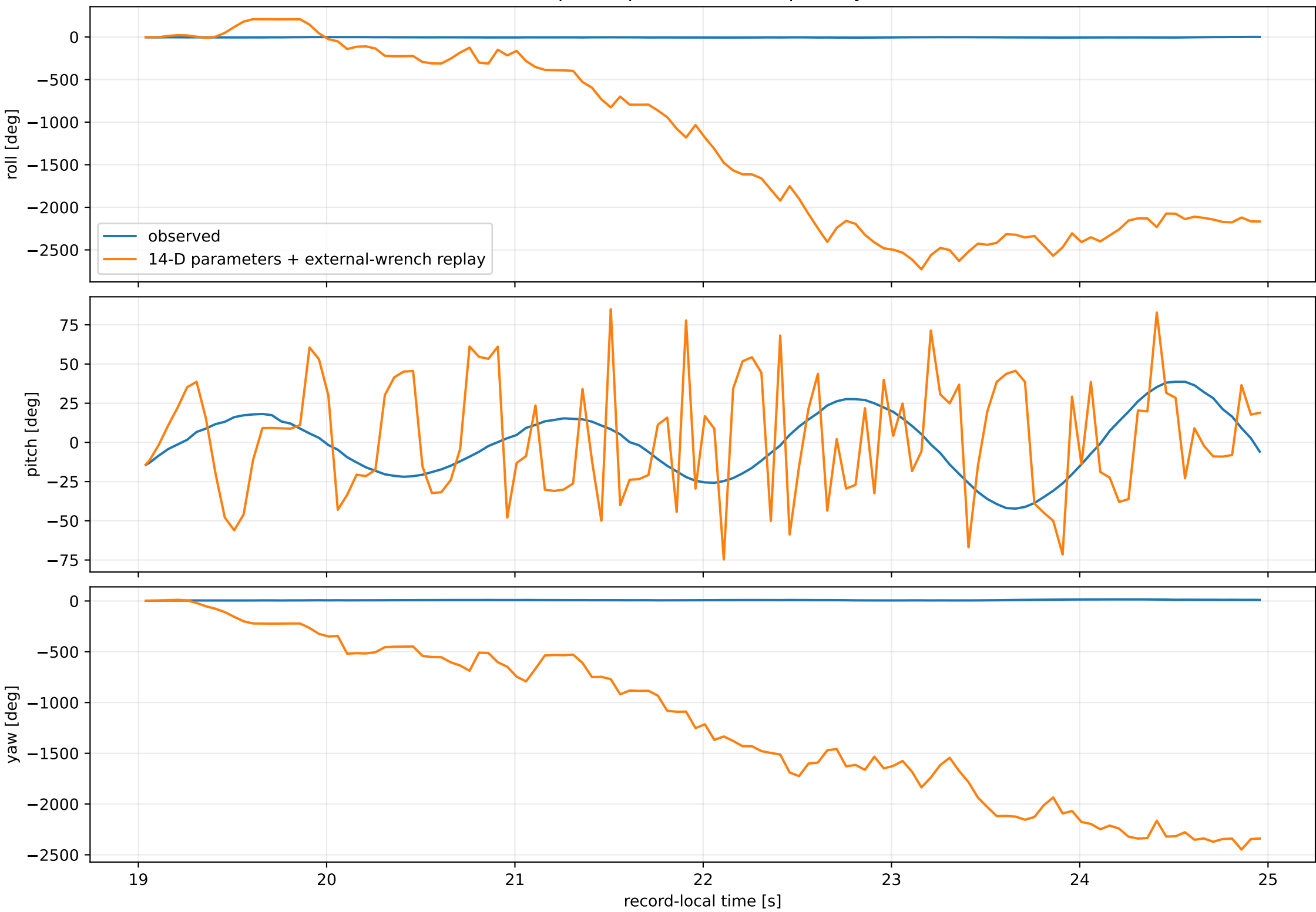


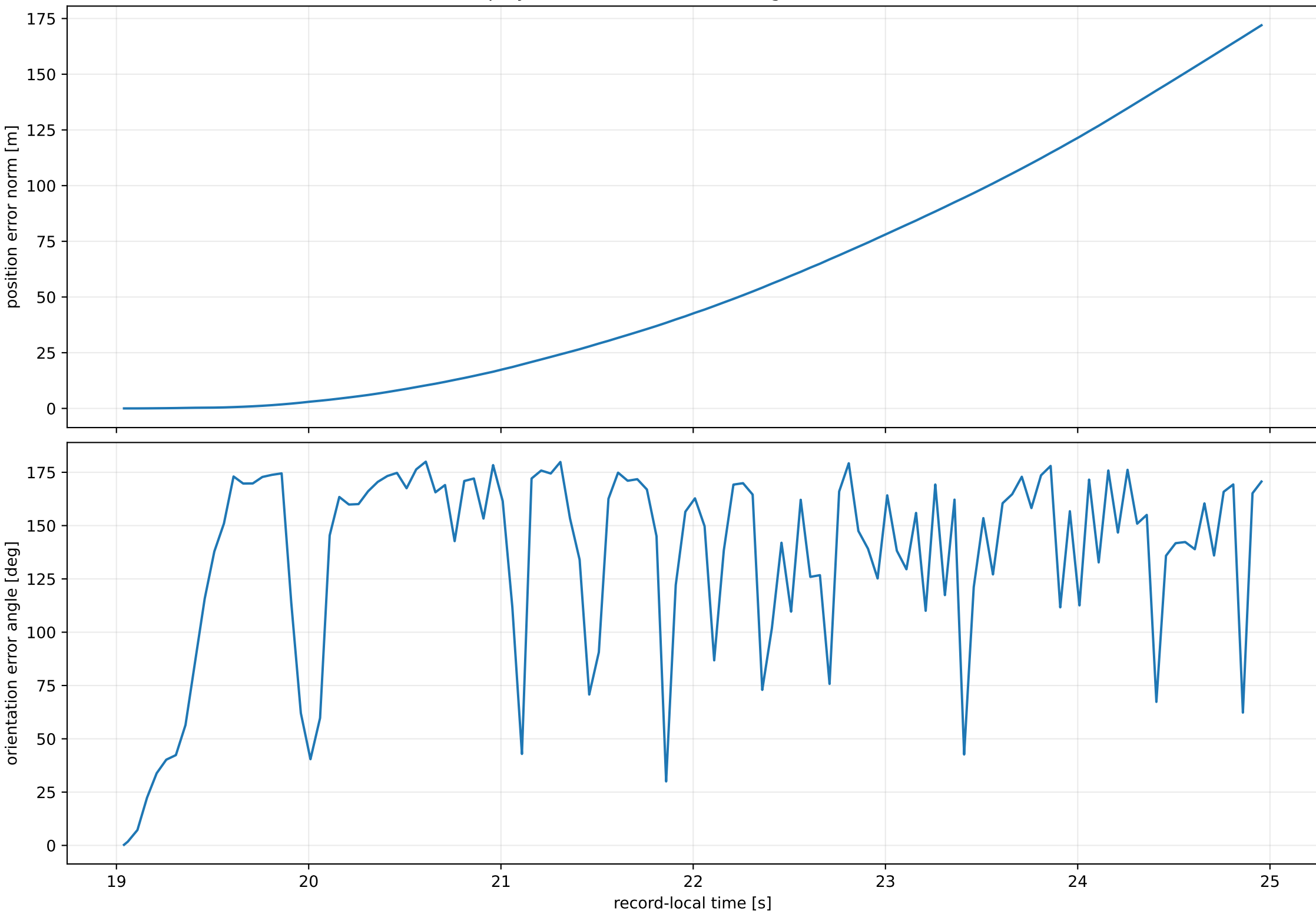
Fine pose reproduction: position



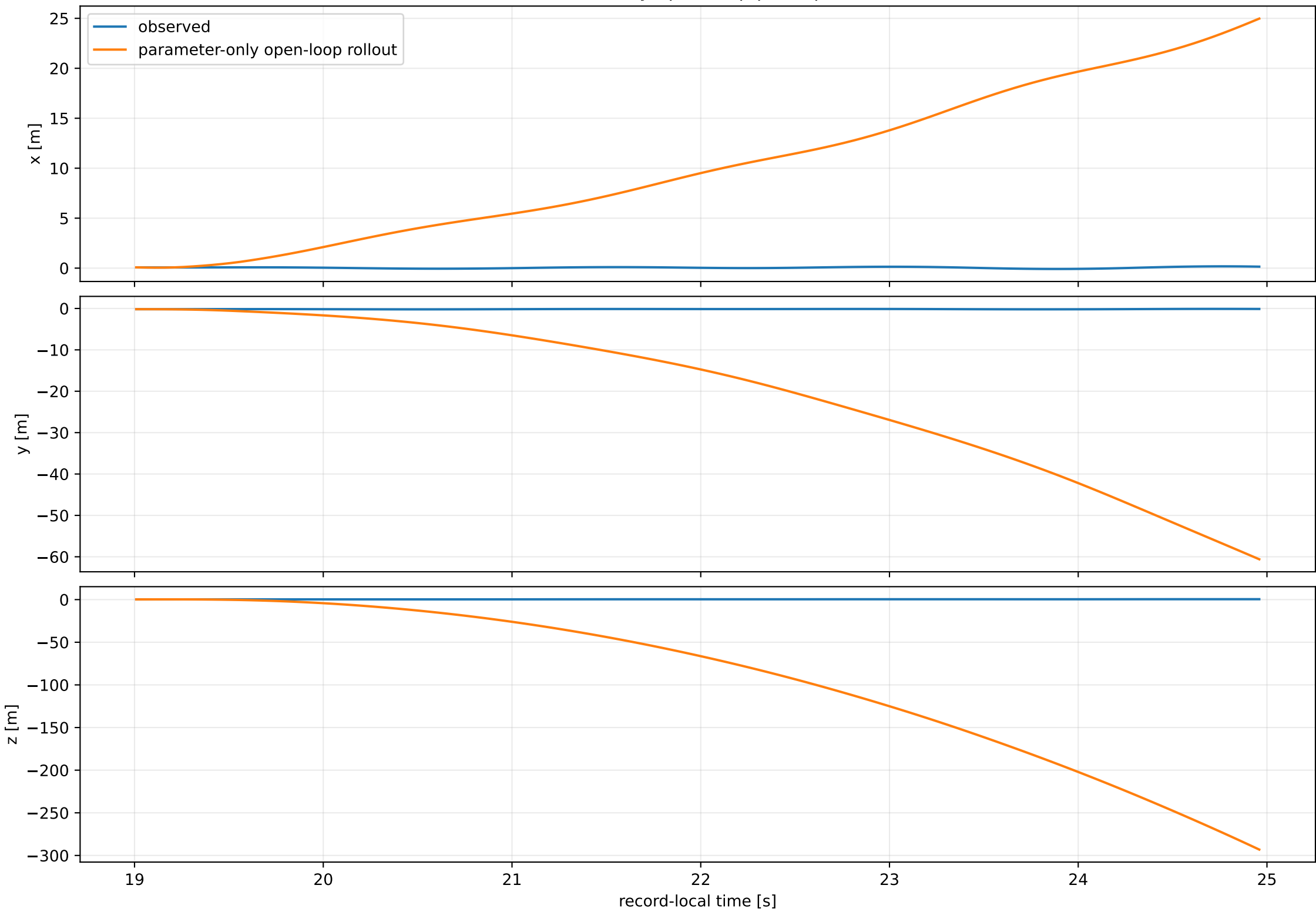
Fine pose reproduction: roll / pitch / yaw



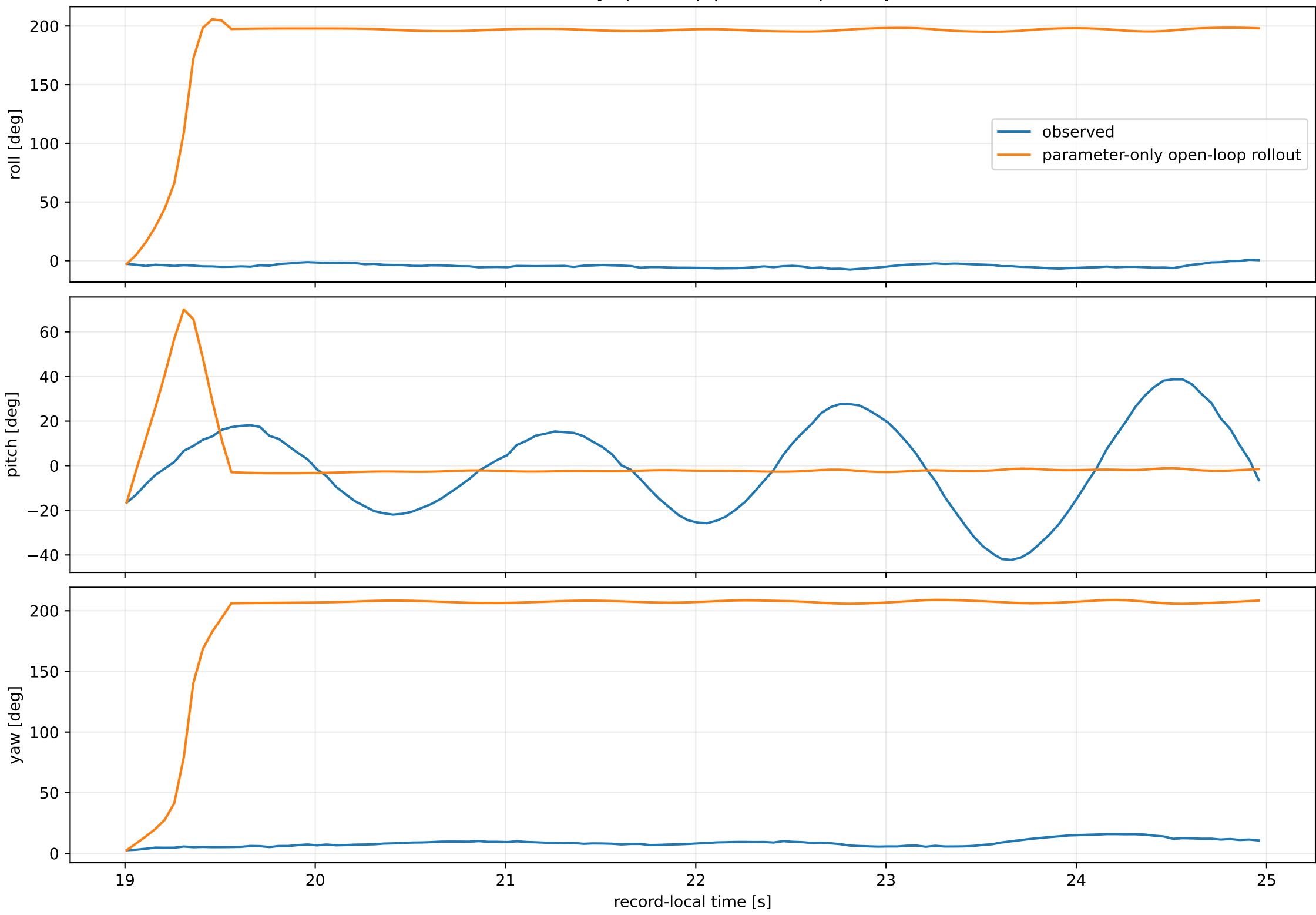
Fine replay error after the 14-D change — lower is better



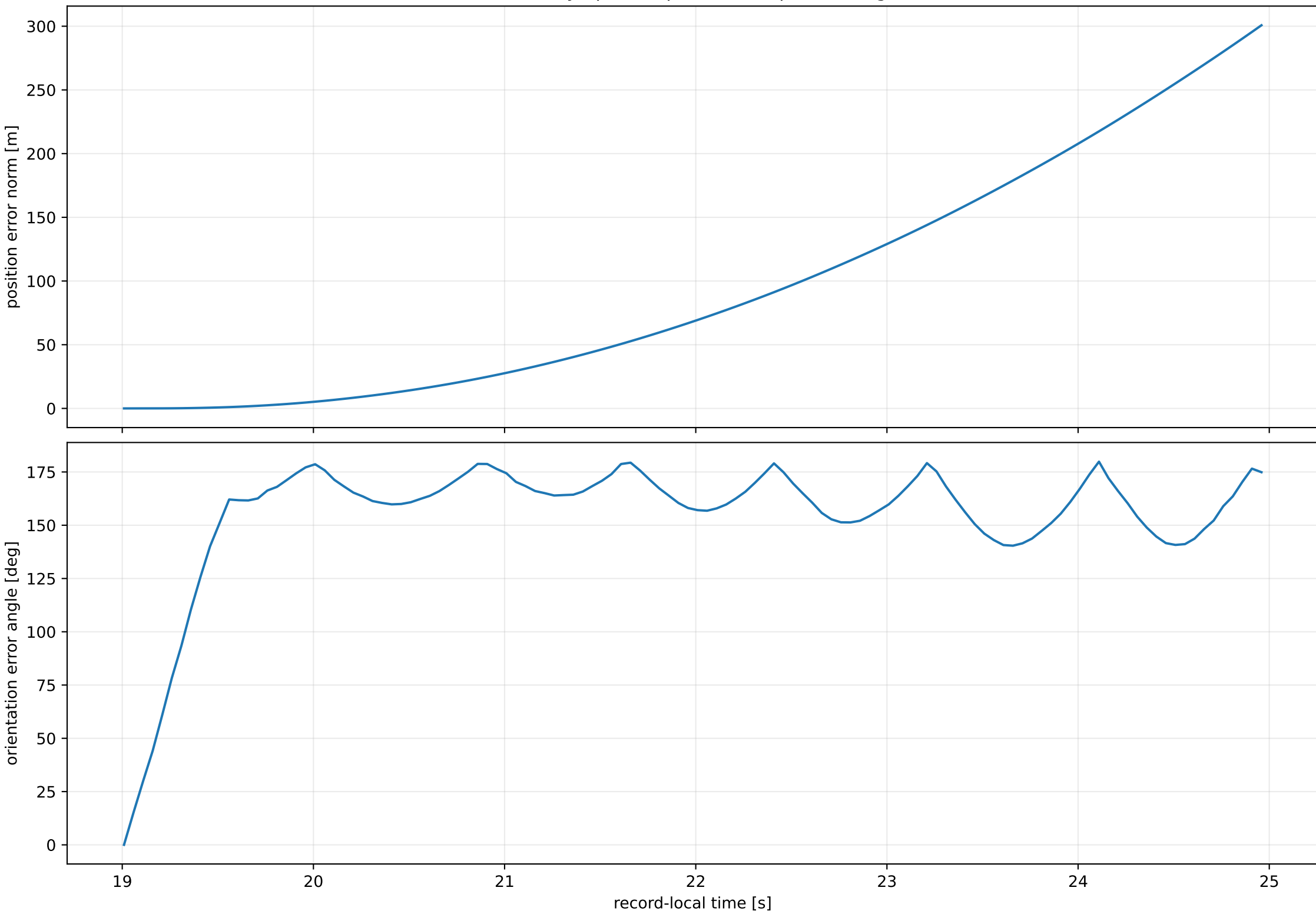
Parameter-only open-loop pose: position



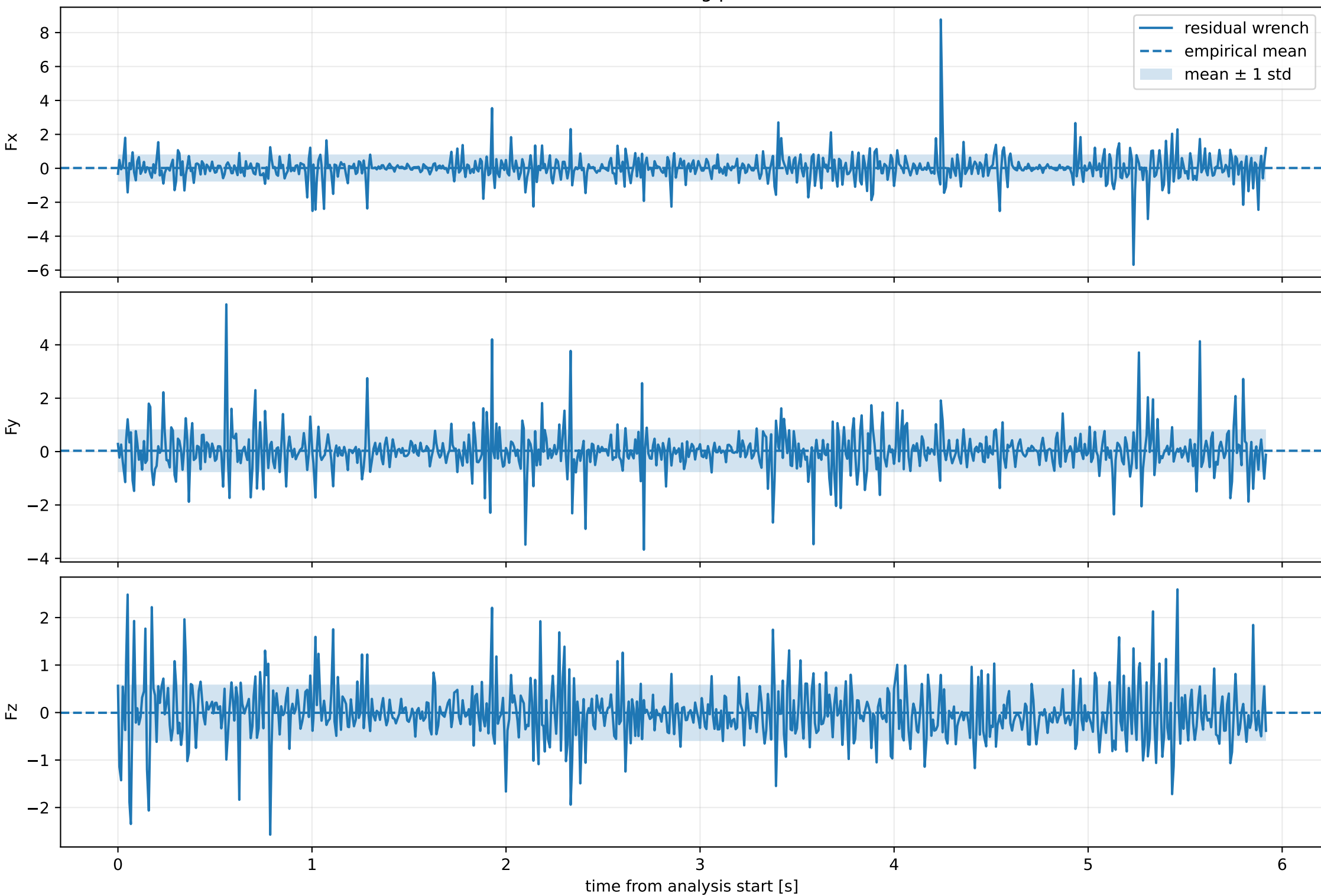
Parameter-only open-loop pose: roll / pitch / yaw



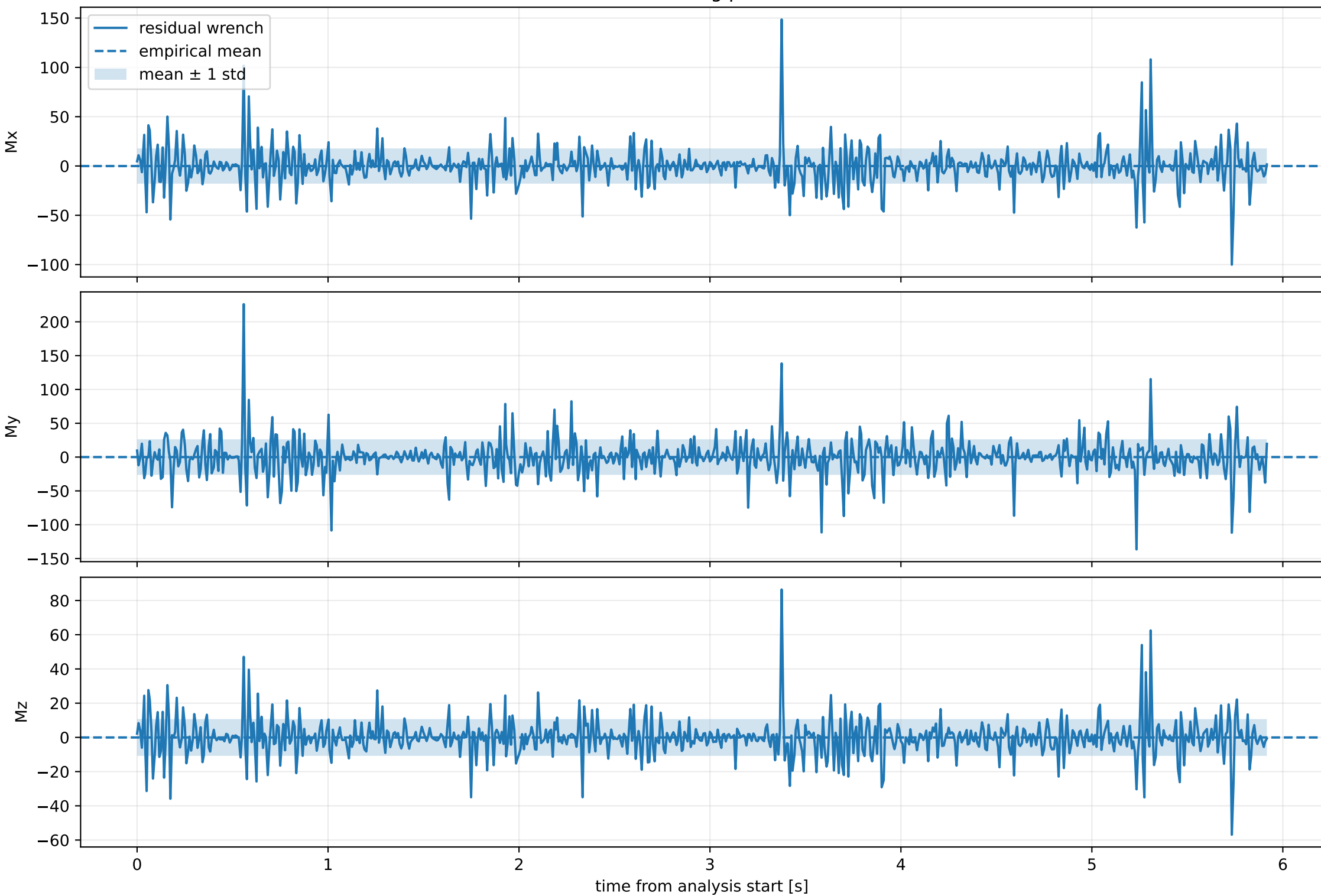
Parameter-only open-loop rollout (separate diagnostic)



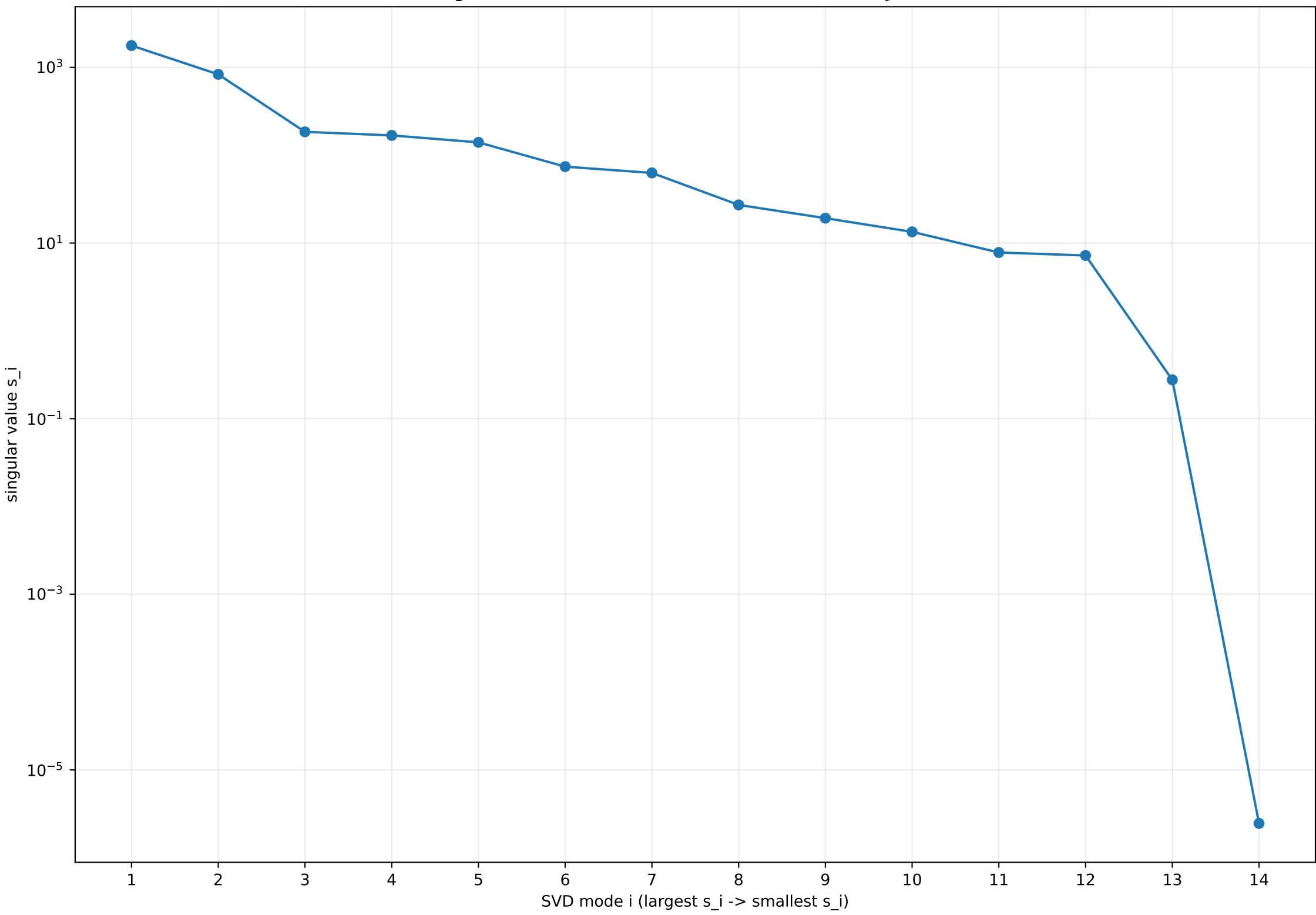
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



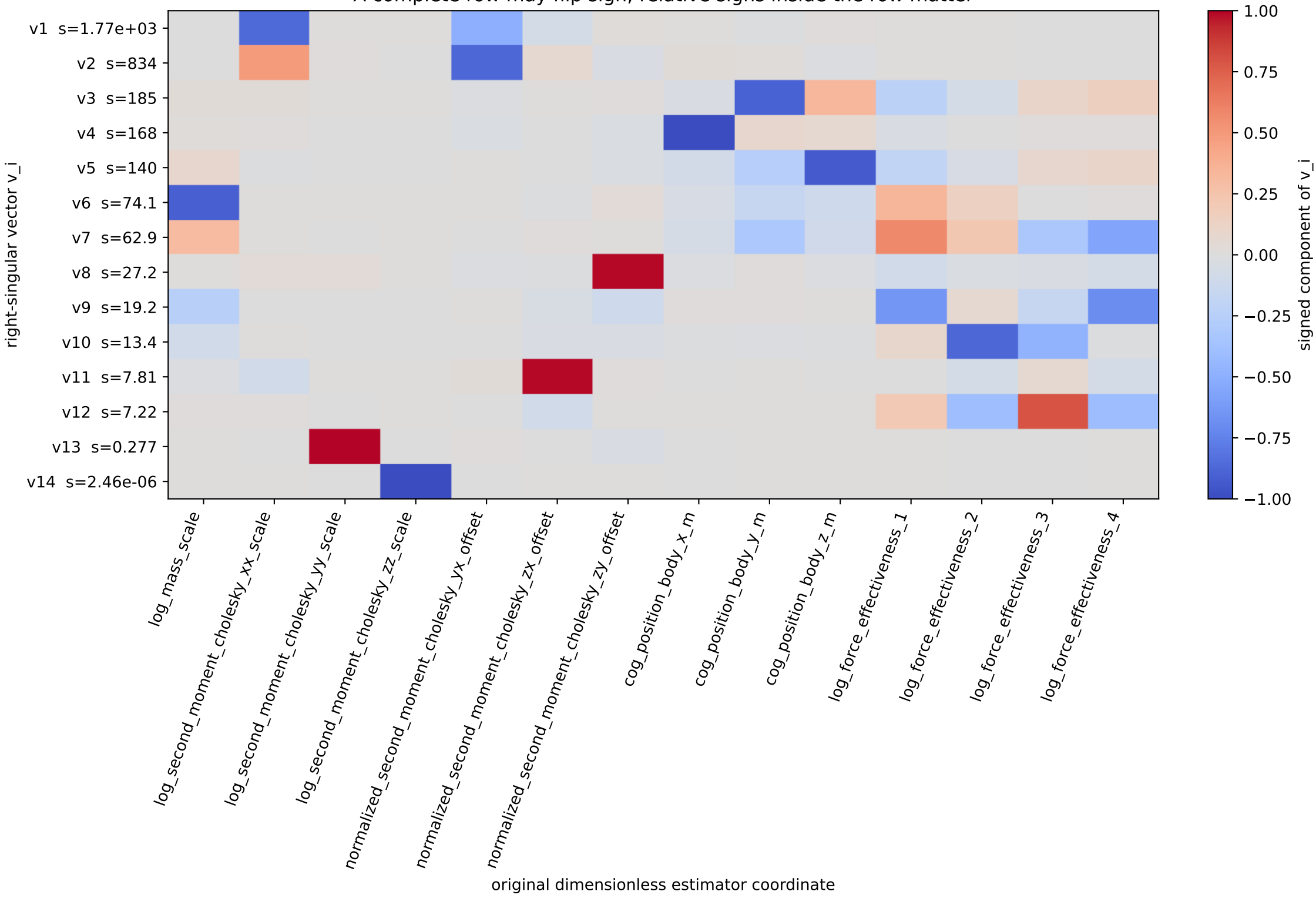
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.55785640129

inertia J [kg m²]

[0.56987186 0.036187791 0.37430917]
[0.036187791 0.82481873 -0.055046061]
[0.37430917 -0.055046061 0.26569847]

principal moments [kg m²]

[0.0084431879, 0.82175135, 0.83019452]

CoG position in body frame [m]

[-0.03363214, -0.041129205, 0.055589754]

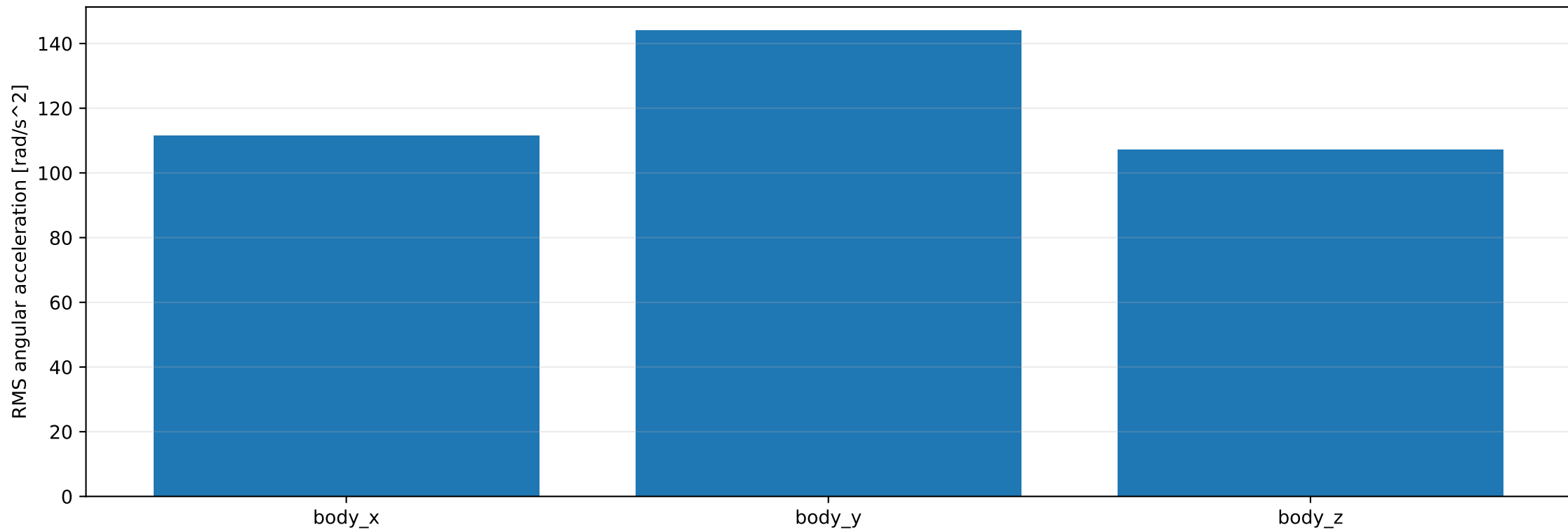
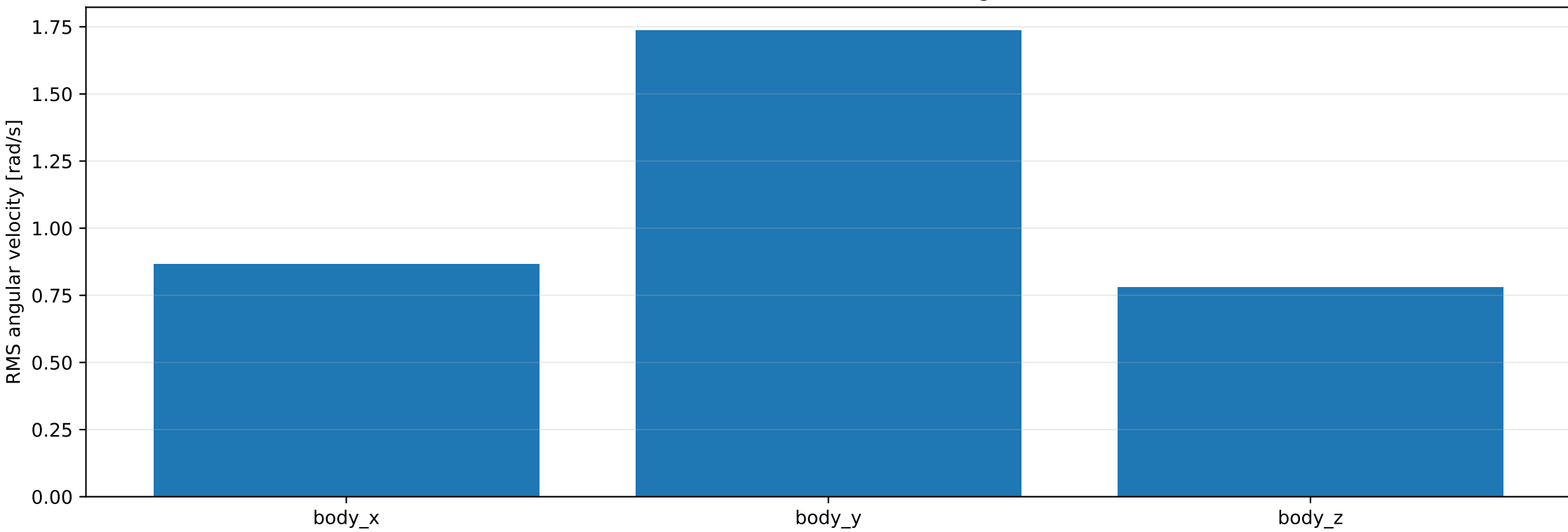
rotor force effectiveness

[1.3075354, 0.75700644, 0.61222153, 1.0477211]

selected delay [s]

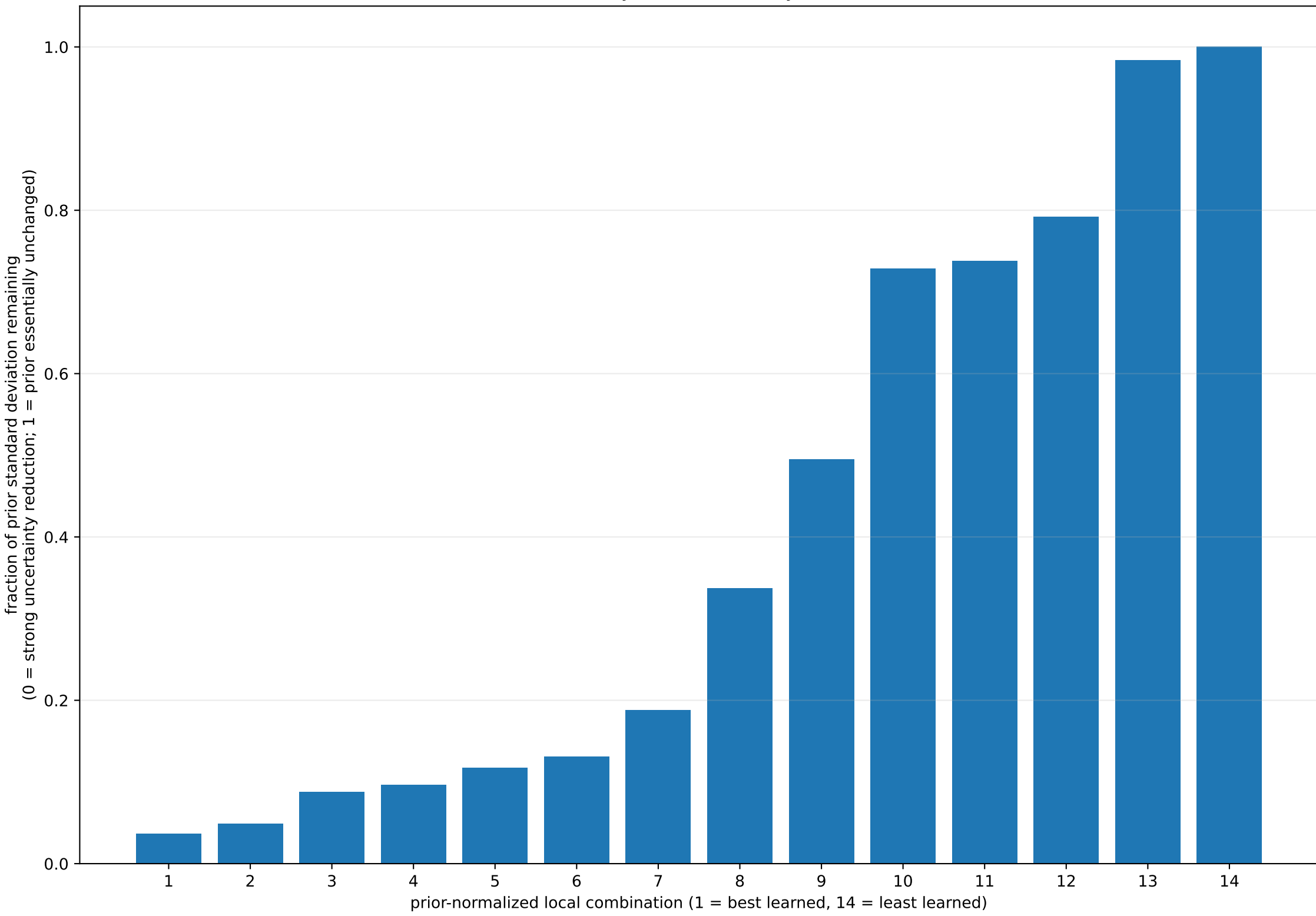
0.152624524376

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

